

Utsav Chaudhary

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Robotics + Embedded Software Engineer. Build VR/teleop interfaces, ROS2 stacks, ESP32 firmware, and real-time controls.

EDUCATION

DREXEL UNIVERSITY

Philadelphia, PA

Bachelor of Science

2026

Major in Computer Science

Cumulative GPA: 3.75/4.0; Dean's List

Relevant Coursework: Machine Learning, Design with Embedded Processors, Concurrent Programming, Probability & Statistics

EXPERIENCE

FAUNA ROBOTICS

New York, NY

Teleoperation Intern

Sep 2025 – Mar 2026

- Utilized Foxglove bridge, WebRTC, and custom REST APIs to connect a ROS2 backend with a React frontend, building a robot status, control, and debug dashboard for live monitoring and teleoperation.
- Designed VR interfaces to switch robot modes/skills and to visualize robot state (e.g., 3D-map, joint pose, temperatures).

UPRIGHT VR

Philadelphia, PA

VR Developer

Sep 2022 – Aug 2024

- Developed gamified VR rehabilitative exercises using UnityXR and C# for elderly patients with vestibular/ocular disabilities.
- Designed a local database to track patient performance data in real time and auto-generate charts.

DREXEL UNIVERSITY

Philadelphia, PA

Student Researcher

Apr 2022 – Sep 2024

- Engineered two-way serial communication between a microcontroller and Unity applications, transmitting up to 64 bytes at 200 Hz to support realtime data transfer for interactive therapy game development.
- Developed customized therapy games and a game controller using microcontrollers, an encoder, and biometric sensors to monitor heart rate and stress, enhancing real-time patient feedback during therapy.

CHIMPVINE

Jersey City, NJ

Game Developer

Sep 2020 – Aug 2021

- Developed 2D e-learning games using Unity and C# for primary-grade students.
- Optimized game performance for mobile browsers, reducing load times by 20% and increasing mobile accessibility.

PROJECTS

HOPE JR – VR-Teleoperation Robot (Presented in Humanoids Summits 2024/2025)

June 2024

- Developed a 3D printed VR-controlled robot using ROS2, WebRTC, UnityXR, ESP32 microcontrollers, and 35 servo motors to achieve 6-DOF end-effector tracking for real-time interaction.
- Engineered an inverse kinematics solver to translate VR controller and headset data into servo motor commands.

HYBRID CLASSROOM – Cross-Platform Virtual Classroom

Dec 2021

- Built a 3D multiplayer virtual classroom using UnityXR and Photon Networking, supporting immersive, realtime interaction for 20+ concurrent users across VR headsets and web browsers (Unity WebGL).
- Implemented WebRTC-based video, audio, and text chat to enable cross-platform communication.
- Designed a cross-platform virtual whiteboard using Unity and Photon Networking.

SKILLS & AWARDS

Technical Skills: ROS2, UnityXR, Unreal Engine, C#, C++, Python, Java, React, JSX/TSX, Arduino, Linux, Git, WebRTC, REST APIs, Foxglove, ESP32, Photon Networking, Embedded Systems, Inverse Kinematics, Real-time Systems, VR/XR Development, CMake

Awards:

- MIT Reality Hack, 1st Place in Healthcare and Startup Track, 3rd Place Overall, 2024
- Microsoft Imagine Cup, Asia Finalist, 2020
- NASA Space Apps Challenge, Global Nominee, 2019